IIT-PATNA-PROJECTS

IIT-Patna-Projects: Engineering Ideas into Real-World Solutions

Engineering Ideas into Real-World Solutions.



Cloud Computing

DevOps

Distributed Systems

Cybersecurity

AI

Software Engineering

THE WORKSPACE

One workspace for building complete working systems

Build the whole system.

A centralized GitHub organization for technology projects, practical implementations, experimentation, and technical learning.

Implement

Connect technologies into working applications.

Experiment

Test system behavior under controlled conditions.

Learn

Capture knowledge from practical engineering.

The emphasis is complete working systems, rather than technologies learned in isolation.

PROJECT OBJECTIVES

Four objectives drive every project

Build practical solutions

Develop software and technology that address real implementation needs.

Explore modern computing

Investigate cloud platforms and emerging computing technologies.

Evaluate system quality

Examine application performance, security, and reliability.

Document the evidence

Record technical knowledge and experimental outcomes.

ENGINEERING SCOPE

Six technical domains, one engineering practice



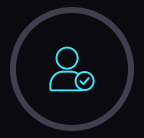
Cloud Computing and Distributed Systems

Scaling, load balancing, serverless



DevOps and Infrastructure Automation

CI/CD, containers, infrastructure as code



Performance Monitoring and Reliability Engineering

Metrics, dashboards, recovery testing



Cybersecurity and Information Security

Access control, least privilege



Artificial Intelligence and Data Engineering

Data pipelines, notebooks, analysis



Software and Application Development

Web apps, APIs, databases

TECHNOLOGY TOOLKIT

A flexible stack, chosen per project

Programming

Python, Java, C, JavaScript, TypeScript, SQL

Frontend

React, HTML, CSS

Backend

FastAPI, REST APIs

Cloud

AWS, Microsoft Azure, Google Cloud

DevOps

Docker, Jenkins, GitHub Actions, Terraform

Monitoring

Prometheus, Grafana, CloudWatch

Databases

PostgreSQL, MySQL, Redis

Data Science

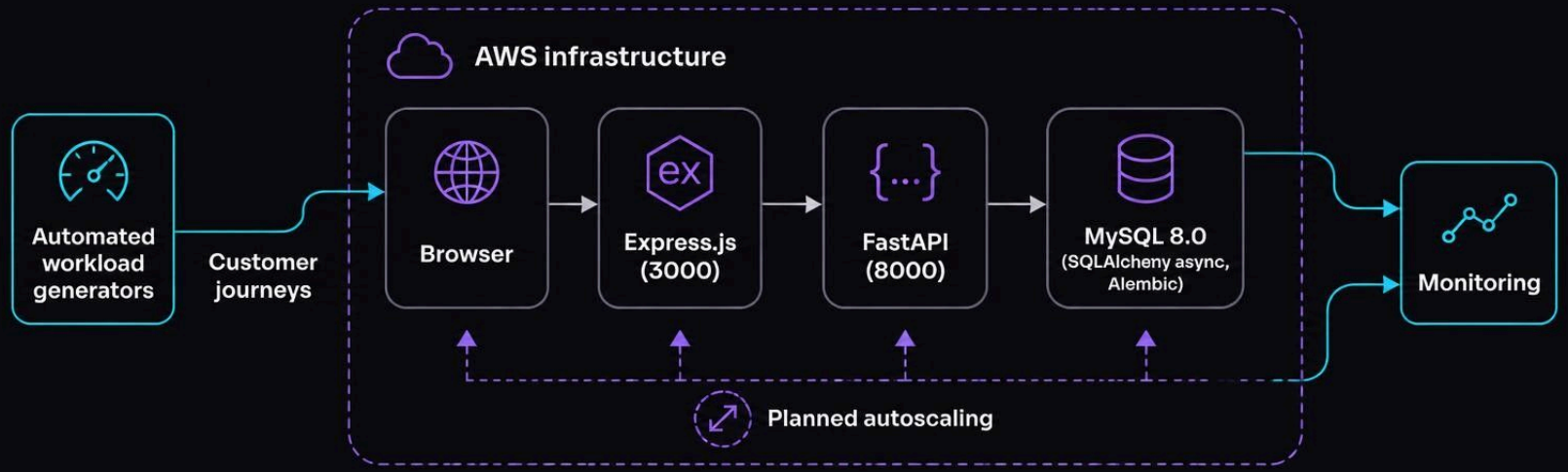
NumPy, pandas, Matplotlib, Jupyter

Technology choices vary by project.

ShopScale: planned e-commerce scalability and reliability

PLANNED / IN DEVELOPMENT

NO COMPLETED DEPLOYMENT CLAIMED



Planned storefront

Product catalog, cart, inventory, simulated checkout

Planned experiments

Automated customer journeys, controlled traffic, performance monitoring

Planned evaluation

SLA evaluation, recovery testing, bot detection

Serverless and cloud-native research: areas of exploration

EXPLORATION · NOT VERIFIED IMPLEMENTATION



Compute

Serverless computing and Function-as-a-Service

Architecture

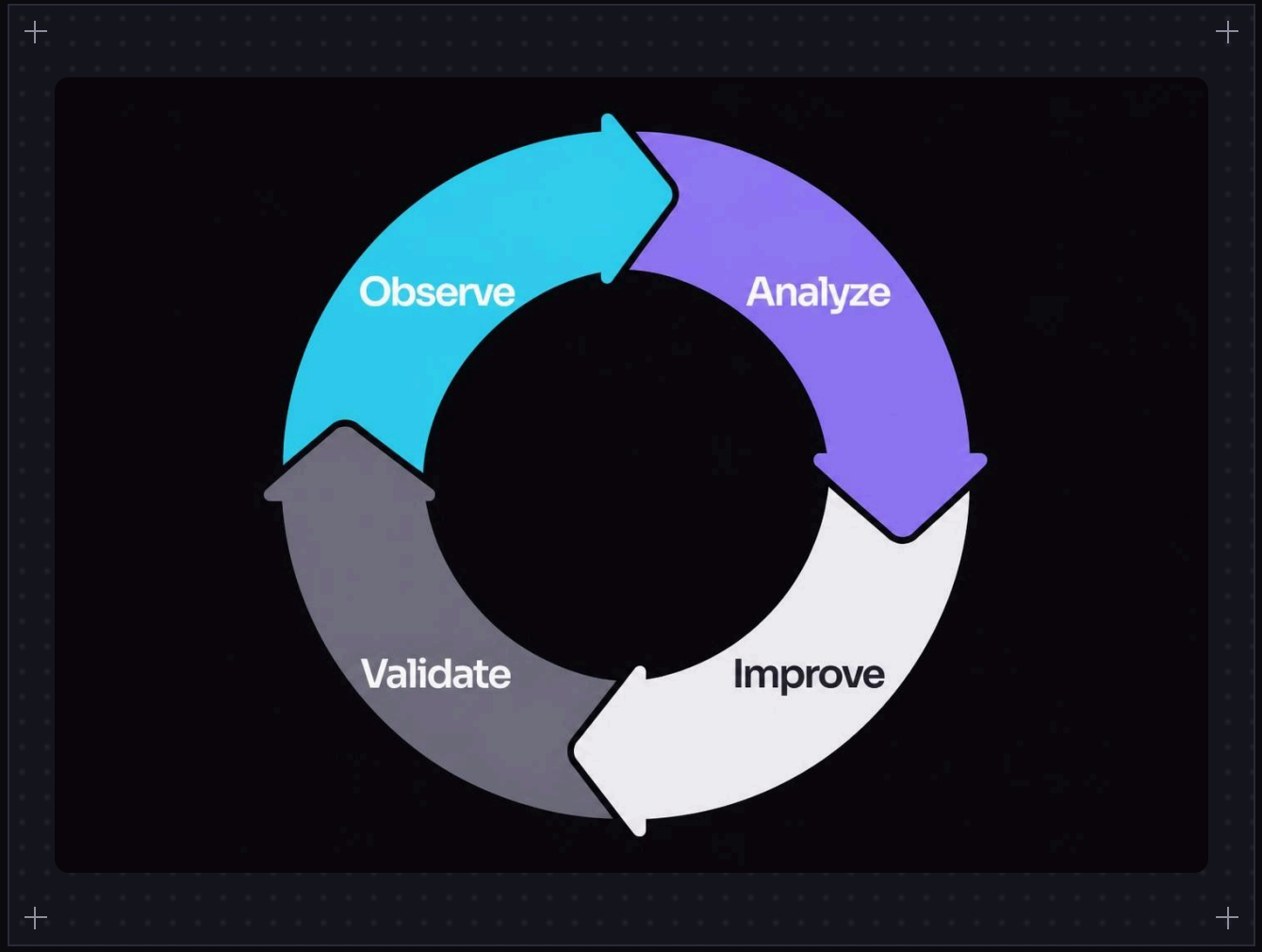
Distributed workflows, hybrid and multi-cloud architectures

Evaluation

Cloud storage and performance benchmarking

SYSTEM QUALITY

Dependable applications need observability and security



Observability

Metrics, logging, resource utilization

Security

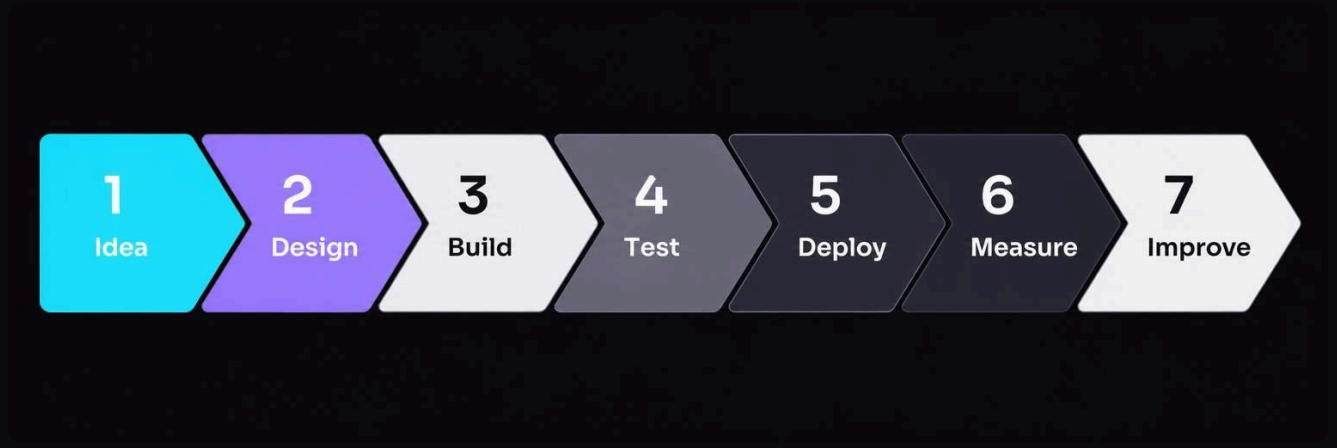
Access control, least privilege

Reliability

Performance testing, incident detection, recovery

ENGINEERING METHOD

Prototype first, then measure everything



- Build functional prototypes before adding complexity.
- Validate assumptions through controlled experiments.
- Measure outcomes instead of making unsupported claims.
- Document architecture, setup, testing, and limitations.
- Collaborate, review, and continuously improve.

Learn by Building.
Grow by Experimenting.

A meaningful collection of practical projects across cloud engineering, DevOps, distributed systems, AI, cybersecurity, and emerging technologies.

From Technical Concepts to Practical Implementations.

Practical projects. Controlled experiments. Documented learning.